DPA PROJECT - PART 1

Group Name - Y2K

Group members -

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Project overview

Analyze how community-level crime varies with unemployment and overall hardship by joining annual Chicago crime records (2019–2023) to the Hardship Index via the Community Area fields present in the crime tables, without introducing external geography layers.
Produce a unified community–year dataset and estimate relationships between incident volumes and Unemployment Rate and Hardship Index

Dataset identification

- Crime data: "Crimes – 2019" "Crimes – 2020," "Crimes – 2021," "Crimes – 2022," and "Crimes – 2023," each an annual table with incident-level fields including Date, Block, Primary Type, Arrest, and Community Area attributes.
- Socioeconomic: Hardship Index CSV with Area Number, Community, Unemployment Rate, Poverty Rate, No HS Diploma Rate, Dependency, Per Capita Income, and a composite Hardship Index Value.

Key attributes

- Join keys: Use the crime tables' Community Area numeric field and/or Community Area name to align to the Hardship CSV's Area Number and Community, avoiding spatial joins.
- Crime variables: Date/Year, Block, Primary Type, Arrest, Domestic, and Community Area fields to enable year filtering, category grouping, and community-level aggregation.

Tools and technologies

- Processing: pandas for cleaning/union, and SQL for schema unification and community–year aggregation prior to modeling.
- Modeling/visuals: scikit-learn for basic regressions/classification and matplotlib/seaborn for plots.

Data preparation and integration

- Annual ingestion: Export or query each annual table (2019–2023) and standardize column names and types so concatenation preserves Date/Year, Block, Primary Type, Arrest, and Community Area fields.
- Union: Concatenate the four annual DataFrames and restrict rows to 2019–2023, ensuring a single tidy incidents table for downstream grouping.

- Community assignment: Rely on the existing Community Area numeric/name fields from the annual crime tables; if missing for some rows, document the omission and exclude those rows to avoid adding external geography sources.
- Hardship merge: Clean the Hardship CSV, keep Area Number, Community, Unemployment Rate, and Hardship Index Value, and create a lookup keyed by Area Number for joining to the crime aggregates.

Aggregation and features

- Community–year table: Group by Community Area and Year to compute total incidents and optional category counts (e.g., violent vs. property or selected top offense types) for 2019–2023.
- Socioeconomic join: Left-join Hardship variables onto the community–year table via Area Number/Community, producing a final dataset for analysis.

Dataset JOIN

- Select columns: From each annual crime dataset, select block and other columns to rename and cast types consistently, since the analysis is by community.
- Union data: Stack the four annual tables (2019–2023 subset you are using) into a single incidents table, keeping only the columns needed for community-level analysis.
- Community linkage: Standardize the crime "address" or BLOCK information to match the community names used in the Hardship Index, then use community_area to join to the Hardship table's Area Number/Community so each incident is assigned to the correct community.
- Aggregate: Group by community area and hardship index features to produce total incident counts and optional category splits for each community, yielding a compact community-level panel.
- Join hardship: Merge the community-level crime table with the Hardship Index on community_area to attach Unemployment Rate, Hardship Index, and related socioeconomic variables to each community's crime totals.

Analysis and modeling

- Descriptive: Plot incident counts by community and year, and compare distributions across high- vs. low-hardship communities to reveal broad associations.
- Regression: Estimate relating incident counts to Unemployment Rate and Hardship Index Value with controls for year fixed effect.

Expected outcomes

- Unified dataset: Community–year panel for 2019–2023 with crime counts and joined unemployment and hardship measures, suitable for modeling and visualization.
- Findings: Clear evidence on whether higher unemployment and hardship correlate with higher incident volumes, presented with interpretable coefficients and plots per the course template.